Zika Virus: Yet Another Emerging Threat to Nepal.

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Zika virus (ZIKV) is a flavivirus with single stranded RNA related to yellow fever, dengue, West Nile, and Japanese encephalitis viruses and is transmitted by Aedes mosquitoes primarily by Aedes aegipti which is widely distributed in Nepal. ZIKV was first identified incidentally in Rhesus monkey in Uganda in 1947 and human infection in 1952; and by now outbreaks of ZIKV disease have been recorded in Africa, the Americas, Asia and the Pacific. The World Health Organization (WHO) has recently declared the ZIKV an international public health emergency. The aim of this paper is to briefly summarize origin, signs, symptoms, transmission, diagnosis, preventions and management of ZIKV and possible threat to Nepal in light of endemicity of other arbovirus infections and common mosquito vector species in Nepal. Keyword: Aedes aegypti; aedes albopictus; zika virus; microcephaly; birth defect; Nepal.

Different populations of Aedes aegypti and Aedes albopictus (Diptera: Culicidae) from Central Africa are susceptible to Zika virus infection.

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Zika virus (ZIKV) is a Flavivirus (Flaviviridae) transmitted to humans mainly by the bite of an infected Aedes mosquitoes. Aedes aegypti is the primary epidemic vector of ZIKV and Ae. albopictus, the secondary one. However, the epidemiological role of both Aedes species in Central Africa where Ae. albopictus was recently introduced is poorly characterized. Field-collected strains of Ae. aegypti and Ae. albopictus from different ecological settings in Central Africa were experimentally infected with a ZIKV strain isolated in West Africa. Mosquitoes were analysed at 14- and 21-days post-exposure. Both Ae. aegypti and Ae. albopictus were able to transmit ZIKV but with higher overall transmission efficiency for Ae. aegypti (57.9%) compared to Ae. albopictus (41.5%). In addition, disseminated infection and transmission rates for both Ae. aegypti and Ae. albopictus varied significantly according to the location where they were sampled from. We conclude that both Ae. aegypti and Ae. albopictus are able to transmit ZIKV and may intervene as active Zika vectors in Central Africa. These findings could contribute to a better understanding of the epidemiological transmission of ZIKV in Central Africa and develop suitable strategy to prevent major ZIKV outbreaks in this region.

Gut symbiont-derived sphingosine modulates vector competence in Aedes mosquitoes.

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The main vectors of Zika virus (ZIKV) and dengue virus (DENV) are Aedes aegypti and Ae. albopictus, with Ae. aegypti being more competent. However, the underlying mechanisms remain unclear. Here, we find Ae. albopictus shows comparable vector competence to ZIKV/DENV with Ae. aegypti by blood-feeding after antibiotic treatment or intrathoracic injection. This suggests that midgut microbiota can influence vector competence. Enterobacter hormaechei_B17 (Eh_B17) is isolated from field-collected Ae. albopictus and conferred resistance to ZIKV/DENV infection in Ae. aegypti after gut-transplantation. Sphingosine, a metabolite secreted by Eh_B17, effectively suppresses ZIKV infection in both Ae. aegypti and cell cultures by blocking viral entry during the fusion step, with an IC50 of approximately 10 μM. A field survey reveals that Eh_B17 preferentially colonizes Ae. albopictus compared to Ae. aegypti. And

field Ae. albopictus positive for Eh_B17 are more resistant to ZIKV infection. These findings underscore the potential of gut symbiotic bacteria, such as Eh_B17, to modulate the arbovirus vector competence of Aedes mosquitoes. As a natural antiviral agent, Eh_B17 holds promise as a potential candidate for blocking ZIKV/DENV transmission.

Inflammatory-dependent Sting activation induces antiviral autophagy to limit zika virus in the Drosophila brain.

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Incidences of congenital syndrome associated with maternal zika virus (ZIKV) infection during pregnancy are well documented; however, the cellular and molecular mechanisms by which ZIKV infection causes these devastating fetal pathologies are still under active investigation. ZIKV is a member of the flavivirus family and is mainly transmitted to human hosts through Aedes mosquito vectors. However, in vivo models for the neurological tropism of the virus and the arthropod vector have been lacking. A recent study published in Cell Host & Microbe from Dr. Sara Cherry's lab investigates both of these key aspects of the ZIKV infectious life cycle. Liu et al. demonstrate how inflammatory activated Sting/dSTING-dependent antiviral macroautophagy/autophagy is sufficient to restrict ZIKV infection in the Drosophila melanogaster brain. Additionally, this study provides further evidence for the ancestral function of autophagy in protecting host cells from viral invaders. Abbreviations: AGO2: Argonaute 2; ATG: autophagy-related; Dcr-2: Dicer-2; DptA/Dipt: Diptericin A; Drs: Drosomycin; DCV: Drosophila C virus; IMD: immune-deficiency; qRT-PCR: quantitative real-time PCR; Rel/NF-κB: Relish; RNAi: RNA interference; ZIKV: zika virus.

Current Zika virus epidemiology and recent epidemics.

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The Zika virus (ZIKV) is a mosquito-borne flavivirus (Aedes), similar to other arboviruses, first identified in Uganda in 1947. Few human cases were reported until 2007, when a Zika outbreak occurred in Yap, Micronesia, even though ZIKV activity had been reported in Africa and in Asia through virological surveillance and entomological studies. French Polynesia has recorded a large outbreak since October 2013. A great number of cases and some with neurological and autoimmune complications have been reported in a context of concurrent circulation of dengue viruses. The clinical presentation is a "dengue-like syndrome". Until the epidemic in French Polynesia, no severe ZIKV disease had been described so far. The diagnosis is confirmed by viral genome detection by genomic amplification (RT-PCR) and viral isolation. These two large outbreaks occurred in a previously unaffected area in less than a decade. They should raise awareness as to the potential for ZIKV to spread especially since this emergent disease is not well known and that some questions remain on potential reservoirs and transmission modes as well as on clinical presentations and complications. ZIKV has the potential to spread to new areas where the Aedes mosquito vector is present and could be a risk for Southern Europe. Strategies for the prevention and control of ZIKV disease should include the use of insect repellent and mosquito vector eradication.

Zolbetuximab for Claudin18.2-positive gastric or gastroesophageal junction cancer.

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Claudins (CLDNs) are a family of major membrane proteins that form components of tight junctions. In normal tissues, CLDNs seal the intercellular space in the epithelial sheets to regulate tissue permeability, paracellular transport, and signal transduction. Claudin18.2 (CLDN18.2), a member of the CLDN family, is expressed specifically in gastric mucosal cells in normal tissue, and its expression is often retained in gastric cancer cells. CLDN18.2 is ectopically expressed in many cancers other than gastric cancer such as esophageal cancer, pancreatic cancer, biliary tract cancer, non-small-cell lung cancer, and ovarian cancer. Structurally, CLDN18.2 is localized on the apical side of the cell membrane and has extracellular loops capable of binding monoclonal antibodies. Upon malignant transformation, CLDN18.2 is exposed to the cell surface of the whole membrane, which enables the binding of monoclonal antibodies. Based on these characteristics, CLDN18.2 was considered to be optimal for target therapy, and zolbetuximab was developed which is a first-in-class chimeric immunoglobulin G1 monoclonal antibody highly specific for CLDN18.2. It binds to CLDN18.2 on the tumor cell surface and stimulates cellular and soluble immune effectors that activate antibody-dependent cytotoxicity and complement-dependent cytotoxicity. Recently, zolbetuximab combined with chemotherapy demonstrated a survival benefit in patients with CLDN18.2-positive and HER-2-negative gastric or gastroesophageal junction cancers in the global phase III SPOTLIGHT and GLOW trials. From these clinically meaningful results, CLDN18.2-targeting therapy including zolbetuximab has attracted a lot of attention. In this review, we summarize the clinical implications of CLDN18.2-positive gastric or GEJ cancer, and CLDN18.2-targeting therapy, mainly for zolbetuximab.

Claudin-18.2 testing and its impact in the therapeutic management of patients with gastric and gastroesophageal adenocarcinomas: A literature review with expert opinion.

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Claudin-18.2 (CLDN18.2) is a member of the tight junction protein family and is a highly selective biomarker with frequent abnormal expression during the occurrence and development of various primary malignant tumors, including gastric cancer (GC) and esophago-gastric junction adenocarcinomas (EGJA). For these reasons, CLDN18.2 has been investigated as a therapeutic target for GC/EGJA malignancies. Recently, zolbetuximab has been proposed as a new standard of care for patients with CLDN18.2-positive, HER2-negative, locally advanced and metastatic GC/EGJA. The use of CLDN18 IHC assays to select patients who might benefit from anti-CLDN18.2 therapy is currently entering clinical practice. In this setting, pathologists play a central role in therapeutic decision-making. Accurate biomarker assessment is essential to ensure the best therapeutic option for patients. In the present review, we provide a comprehensive overview of available evidence on CLDN18.2 testing and its impact on the therapeutic management of patients with GC/EGJA, as well as some practical suggestions for CLDN18.2 staining interpretation and potential pitfalls in the real-world setting.

Current and Future Biomarkers in Esophagogastric Adenocarcinoma.

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Biomarker-based therapies have shown improved patient outcomes across various cancer types. The purpose of this review to summarize our knowledge of current and future biomarkers in esophagogastric adenocarcinoma (EGA). In this publication, we will review current standard biomarkers in patients with upper GI cancers. We will also discuss novel biomarkers that are under investigations and their associated therapies that are currently in clinical trials. EGAa are a group of heterogeneous diseases, both anatomically and molecularly. There are several established biomarkers (HER2, PD-L1, microsattelite instability or mismatch repair protein expression) that allow for individualized treatments for patients with these cancers. There are also several emerging biomarkers for EGA, some of which have clinically relevant associated therapies. Claudin 18.2 is the furthest along among these. Anti-claudin antibody, zolbetuximab, improved overall survival in biomarker select patients with advanced GEA in two phase 3 studies. Other novel biomarkers, such as FGFR2b and DKN01, are also in the process of validation, and treatments based on the presence of these biomarkers are currently in clinical studies. Ongoing efforts to identify novel biomarkers in EGA have led to enhanced subclassification of upper GI cancers. These advances, coupled with the strategic application of targeted therapies and immunotherapy when appropriate, hold promise to further improve patients outcomes.

Top Gastric Cancer Articles from 2022 and 2023 to Inform Your Cancer Practice.

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The multimodality management of patients with gastroesophageal cancers is rapidly evolving, with the introduction of new therapies against potential molecular targets paving the way to personalized medicine for patients with both resectable and metastatic disease. Over the past 2 years, several important studies evaluating these new targeted therapies, as well as minimally invasive surgical approaches to gastric cancer, have been published. This review article summarizes the top studies published in gastric cancer over the past 2 years that are fundamentally changing our practice approach to gastric cancer patients. First, the long-term safety and efficacy of laparoscopic distal gastrectomy as compared with open gastrectomy for locally advanced gastric cancer was confirmed with the publication of the 5-year outcomes of the CLASS-01 and KLASS-02 randomized clinical trials. In addition, several important studies of perioperative immunotherapy for patients with resectable gastric or gastroesophageal junction cancers are ongoing, and in 2022, an interim analysis of the DANTE trial and the final results of the GERCOR NEONIPIGA study were reported. Lastly, the KEYNOTE-859 and SPOTLIGHT trials address an unmet need for additional targeted therapies for patients with previously untreated, human epidermal growth factor receptor-2 (HER2)-negative, unresectable or metastatic gastroesophageal cancers, incorporating immune checkpoint inhibitors and targeting Claudin-18 isoform 2 (CLDN18.2) with the monoclonal antibody zolbetuximab, respectively. This article summarizes the findings and implications of several important studies published over the past 2 years that are fundamentally changing the way we treat patients with gastroesophageal cancer.

Evidence to Date on the Therapeutic Potential of Zolbetuximab in Advanced Gastroesophageal Adenocarcinoma.

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Gastric adenocarcinoma (GAC) continues to be a prevalent worldwide malignancy and a leading cause of cancer death, and it is frequently cited as incurable. Targeted therapy in GAC has lagged behind other solid tumors. The human epidermal growth factor receptor-2 (HER-2) represented the single target in GACs for many years, seen in approximately 20% of patients with advanced GAC. Recent advances in management now include the addition of immunotherapy checkpoint inhibition to select front-line advanced GACs. Unfortunately, outcomes remain poor for most patients. We anticipate finding a key to future discoveries in GACs in next-generation sequencing and more targeted approaches. Claudin 18.2 (CLDN18.2) has emerged as a therapeutic target in GACs. CLDN18.2 is reportedly expressed in 14-87% of GACs, and CLDN18.2 is available for monoclonal antibody (mAb) binding as it is expressed on the outer cell membrane. Here, we review the exploration of CLDN18.2 as a target in GACs via the use of zolbetuximab (IMAB362). Zolbetuximab is now under priority FDA review for GACs, and we eagerly await the review outcome.